# Nithik Sudeesh

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 8.52 |
| Bharatiya Vidya Bhavan | Class XII (CBSE) | 2019 | 82.00% |
| Bharatiya Vidya Bhavan | Class X (CBSE) | 2017 | 9.40% |

## EXPERIENCE

**Graduate Apprentice - Gaganyaan Project, Vikram Sarabhai Space Centre (VSSC), ISRO**

- Generated 3D data for CM and CES components, supporting engineering drawings and interface verification.
- Assisted with mass-property calculations, drawing reviews, engineering corrections and meeting minutes.
- Coordinated Avionic Deck acceptance testing and participated in structural qualification and integration trials.
- Supported flight-fastener logistics and movement of hardware, avionics packages and technical documents.
- Assisted in technical documentation and fabrication of components for human-spaceflight applications.

**Indian Railways, Coaching Depot & Diesel Loco Shed, Ernakulam, Kerala - One-week internship**

- Exposure to coach maintenance, locomotive maintenance and the transition from diesel to electric traction.

## PROJECTS

**M.Tech Thesis (Ongoing) - Development of Machine-Learning Interatomic Potentials for Refractory Binary Alloys**

- Benchmarking 4 interatomic-potential families - EAM, MEAM, ACE and MACE - in LAMMPS for refractory binary alloys.
- Evaluating lattice parameters, cohesive energies, model accuracy, computational efficiency and system-size scalability across different atomic systems and computing configurations.
- Developing Python workflows for simulation-data processing, quantitative performance benchmarking and visualization of calculated material properties across multiple interatomic potentials and system sizes.

**B.Tech Project - Thermophysical Property Estimation of Anisotropic Structures Using Inverse Heat Transfer | VSSC / ISRO**

- Simulated an aluminium-honeycomb sandwich structure in ANSYS using temperature data from 24 thermocouples.
- Benchmarked Genetic Algorithm, Differential Evolution and Ant Colony Optimizer in MATLAB over 300-1000 K, obtaining mean conductivity errors of 0.18%, 0.12% and 0.39%, respectively, confirming Differential Evolution as most accurate.
- Used Differential Evolution with inverse heat transfer to estimate three conductivities, heat capacity and contact resistance.
- Converged within 60 generations, predicting through-thickness conductivity and heat capacity with 1.1% experimental error.

**B.Tech Minor Project - American Hand-Sign Detection Using YOLOv8**

- Developed a real-time American hand-sign detection system using YOLOv8, trained for 150 epochs on 1,728 annotated images across 26 alphabet classes for efficient recognition in real-time webcam-based applications under varying backgrounds.
- Achieved 92.3% precision, 88.4% recall, 96.5% mAP@50 and 92.4% mAP@50-95 on the validation dataset.
- Attained 45 FPS and outperformed EfficientDet and Faster R-CNN in both accuracy and inference speed.

## SKILLS

**Atomistic Simulation:** LAMMPS; Molecular Dynamics; structure relaxation, cohesive-energy and elastic-property analysis

**Interatomic Potentials:** EAM, MEAM, ACE, MACE - benchmarking and validation

**Machine Learning:** PyTorch, scikit-learn, YOLOv8, Neural Networks, model training and validation

**Scientific Computing:** Python, NumPy, Pandas, Matplotlib, MATLAB, Jupyter Notebook

**Numerical / Engineering:** Inverse heat transfer, optimization, ANSYS Fluent, SolidWorks, AutoCAD; DFT fundamentals

**High Performance Computing:** Linux, Bash and SLURM for running LAMMPS simulations on HPC clusters.

## SCHOLASTIC ACHIEVEMENTS AND TEACHING

- Recipient of the Ministry of Education, Government of India Post-graduate M.Tech fellowship at IIT Hyderabad.
- Awarded Best Project for the B.Tech inverse heat-transfer property-estimation project.
- B.Tech in Mechanical Engineering with Honours in Thermal and Fluids and a Minor in Machine Learning.
- Teaching Assistant for Fundamentals of Scientific Computing and Computational Methods in Materials Science, supporting tutorials, assignment evaluation and student guidance.

## RELEVANT COURSES

Computational Methods in Materials Science, Fundamentals of Scientific Computing, Electronic Structure and Atomistic Modeling of Materials, Machine Learning and Data Analytics in Materials Science, Role of Microstructure in Materials Selection, High Entropy Materials

## ADDITIONAL INFORMATION

Certificate: Google AI Professional Certificate (Coursera)

Languages: Fluent : Malayalam, English | Elementary : Sanskrit, Hindi, Tamil

Interests: Reading, Meditation, Badminton, Swimming and Cycling